

EDUCATION

Barnard College of Columbia University *Sep 2018 – May 2022*
B.A. in Neuroscience & Behavior (Cellular); Minor: Mathematics; GPA: 3.91/4.0, Cum Laude.
Icahn School of Medicine, Neuroscience PhD Program *Aug 2024 – Present*
Mentor: Rita Goldstein, PhD; GPA: 3.94/4.0; NRSA TL1 scholar
University of Washington eScience Institute *Aug 2025*
Summer Program in Neuroimaging and Data Science

RESEARCH EXPERIENCE

Neuropsychomaging of Addiction and Related Conditions Research Program, *Aug 2020 – Present*
Icahn School of Medicine at Mount Sinai

PhD Student (*Aug 2024 – Present*)

- Exploring shared neural responses to drug-themed movies as a naturalistic measure of drug-biased processing, as a function of treatment status and hormonal fluctuations.
- Studying predictive validity of behavioral markers in heroin use disorders, as compared with traditional self-reported severity measures.
- Investigated functional connectivity during extinction of drug memories under methylphenidate in individuals with cocaine use disorder.

Clinical Research Coordinator (*Jul 2022 – Aug 2024*)

- Ran neuroimaging protocols (MRI, PET/MR, EEG), conducted all participant-facing study procedures (recruitment, screening, behavioral tasks, psychometric surveys, interventions), and coordinated clinical procedures (neurology exams, carotid ultrasounds, cardiovascular screening, reviewing blood results).
- Investigated behavioral measures as predictors of prospective treatment adherence as compared to self-reported severity metrics in individuals with heroin use disorder with postdoctoral fellow Ahmet Ceceli, Ph.D.
- Examined recovery of prefrontal cortical-habenular tract impairments with treatment of heroin use disorder with graduate student Sarah King.
- Coordinated and assisted with grant submissions, annual reporting to the NIH, and IRB correspondence.

Research Assistant (*Aug 2020 – Jul 2022*)

- Investigated spoken recall of a drug movie as a naturalistic measure of impaired salience attribution in individuals with heroin use disorder with postdoctoral fellow Ahmet Ceceli, Ph.D.
- Conducted the participant-facing procedures as listed above.

Summer Research Institute, Barnard College *May 2021 – September 2021*

- Received summer funding and supportive programming to investigate spoken recall of a drug movie as an indicator of attentional bias in a naturalistic context in individuals with heroin use disorder.

Losonczy Lab, Mortimer B. Zuckerman Mind Brain Behavior Institute of Columbia University *May 2019 – Jan 2021*

Research Assistant

- Investigated place cell tuning in the mouse hippocampus and its role in spatial memory by examining dendritic membrane potential dynamics using voltage imaging and local field potential recording.
 - Performed neuron segmentations and morphology classification, *in vivo* electroporation of CA1 pyramidal cells under two-photon microscopy guidance, and plasmid preparation.
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AWARDS & RECOGNITIONS

New Investigator Award, American Society of Clinical Psychopharmacology (\$1500)	2026
NRSA TL1 Pre-Doctoral Career Development Training Award	2025-2026
Mount Sinai Trainee Innovation Award (\$600)	2025
Prize for Distinguished Achievement in Neuroscience, Barnard College	2022
Summer Research Institute Grant, Barnard College (\$5,000)	2021
Summer Research Stipend, Losonczy Lab, Zuckerman Institute (\$4,000)	2020
Dean's List, Barnard College	<i>Fall 2018, Fall 2019, Fall 2020-Spring 2022</i>

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Ceceli AO, King SG, **McClain N**, Alia-Klein N, Goldstein RZ. (2023) The Neural Signature of Impaired Inhibitory Control in Individuals with Heroin Use Disorder. *Journal of Neuroscience*, 43(1), 173-182.
2. Ceceli AO, Huang Y, Kronberg G, Malaker P, Miller P, King SG, Gaudreault PO, **McClain N**, Gabay L, Vasa D, Newcorn JH, Ekin D, Alia-Klein N, Goldstein RZ. (2023) Common and distinct fronto-striatal volumetric changes in heroin and cocaine use disorders. *Brain*, 146(4), 1662–1671. (Recognized in [Faculty Opinion](#) as exceptional new finding; Recognized in [Brain](#) editorial)
3. Ceceli AO, Huang Y, Gaudreault PO, **McClain NE**, King SG, Kronberg G, Brackett A, Hoberman GN, Gray JH, Garland EL, Alia-Klein N, Goldstein RZ. (2024). Recovery of inhibitory control prefrontal cortex function in inpatients with heroin use disorder: A 15-week longitudinal fMRI study. *Nature Mental Health*, 2(6), 694-702.
4. Liao Z, Gonzalez KC, Li DM, Yan CM, **McClain NE**, Zhang G, Evans SW, Chavarha M, Simko J, Makinson CD, Lin MZ, Losonczy A, Negrean A. (2024) Functional architecture of intracellular oscillations in hippocampal dendrites. *Nature Communications*, 15(1).
5. Kronberg G*, Ceceli AO*, Huang Y, Gaudreault PO, King SG, **McClain N**, Alia-Klein N, Goldstein RZ. (2024) Naturalistic drug cue reactivity in heroin use disorder: orbitofrontal synchronization as a marker of craving and recovery. *Brain*, awae369.
6. Huang Y, Butelman E, Ceceli AO, Kronberg G, King SG, **McClain NE**, Wong Y, Boros M, Drury K, Sinha R, Alia-Klein N, Goldstein RZ. (2025) Sex and hormonal effects on drug cue-reactivity and its regulation in human addiction. *Biological Psychiatry*.
7. Ceceli AO, King S, Drury K, **McClain N**, Gray J, Dassanayake PS, Newcorn J, Schiller D, Alia-Klein N, Goldstein R. (2025) The neural signature of methylphenidate- and reconsolidation-enhanced memory disruption in human drug addiction. *PNAS*.
8. Ceceli AO, Huang Y, Kronberg G, **McClain N**, King S, Butelman E, Alia-Klein N, & Goldstein RV. (2025) The impaired Response Inhibition and Salience Attribution (iRISA) model of drug addiction: recent neuroimaging evidence and future directions. *Annual Reviews Psychology*.
9. **McClain N***, Ceceli AO*, Kronberg G, Alia-Klein N, Goldstein RZ. (2025) Moving beyond self-report in characterizing drug addiction: Using behavioral measures to inform treatment adherence. *Biological Psychiatry: Global Open Science*. (*equal contribution)

Preprints

1. King S, Kronberg G, **McClain N**, Zhang Z, Ceceli A, Huang Y, Alia-Klein N, Goldstein R. (Under review in *PNAS*). Verbal recall of a drug-themed movie captures treatment effects and neural encoding in heroin use disorder.
2. Kronberg G*, **McClain N***, Hennings AC, Hasson U, Alia-Klein N, Norman K, Goldstein R. (In preparation). Brain-to-brain movie neurofeedback reduces craving in opioid use disorder (*equal contribution)
3. **McClain N**, Kronberg G, Alia-Klein N, Goldstein R. (In preparation) Menstrual phase and hormonal effects on drug-biased synchronization in women with cocaine addiction.
4. **McClain N**, Kronberg G, Alia-Klein N, Goldstein R. (In preparation) Drug-biased brain state dynamics during naturalistic drug cue exposure in individuals with opioid use disorder.

PRESENTATIONS

Oral Presentations

1. **McClain N** & Ceceli AO. (2022) Impaired Salience Attribution in Drug Addiction. Barnard College Neuroscience & Behavior Senior Thesis Conference; New York, NY, 2022.
2. King S, Kronberg G, **McClain N**, Zhang N, Ceceli A, Gray J, Alia-Klein N, Goldstein RZ. (2023) Verbal movie recall reveals heightened self-reference in individuals with heroin use disorder. Society for Neuroscience 2023; Washington, DC. (Also a poster presentation)

Published Abstracts

1. Kronberg, G., Ceceli, A. O., **McClain, N.**, Vasa, D., Malaker, P., Alia-Klein, N., & Goldstein, R. Z. (2021). Inter-Subject Correlations of fMRI Responses to Cinematic Heroin Use in Individuals with Heroin Use Disorder. *Neuropsychopharmacology*, 47, pp. 425-426. (Presented at American College of Neuropsychopharmacology 2021; San Juan, PR and Society for Neuroscience 2021; Virtual)
2. Ceceli AO, King S, Kronberg G, **McClain N**, Vasa D, Alia-Klein N, Goldstein RZ. Speech and viewing choice as naturalistic markers of drug-biased salience attribution in individuals with heroin use disorder. *Neuropsychopharmacology*, 46(1), pp. 410. (Presented at American College of Neuropsychopharmacology 2022; Phoenix, AZ).

3. Kronberg G, Ceceli AO, Huang Y, Gaudreault PO, **McClain N**, Vasa D, Malaker P, Ekin D, Alia-Klein N, Goldstein RZ. (2022). P37. Effects of Abstinence During Treatment on Shared Nucleus Accumbens Reactivity and Craving in Response to a Drug-Related Movie in Heroin Addiction. *Biological Psychiatry*, 91(9), S102-S103. (Presented at Society of Biological Psychiatry 2022; New Orleans, LA)

Poster Presentations

1. **McClain N** & Ceceli AO. Investigating impaired salience attribution in drug addiction via naturalistic stimuli: Speech encoding for drug- and non-drug-related movie scenes in individuals with heroin addiction. Barnard College Lida Orzeck '68 Poster Session 2021; New York, NY.

2. Ceceli AO, King S, **McClain N**, Gaudreault PO, Alia-Klein N, Goldstein RZ. The neural signature of impaired inhibitory control and its recovery in heroin addiction. Gordon Research Conference 2022; Orlando, FL.

3. King S, Kronberg G, **McClain N**, Ceceli A, Gray J, Alia-Klein N, Goldstein RZ. Verbal movie recall reveals heightened self-reference and contextual variability in heroin-addicted individuals. Columbia University Curiosity, Creativity, and Complexity Conference 2023; New York, NY.

4. Huang Y, Ceceli AO, Kronberg G, King S, Brackett A, Hoberman GH, **McClain NE**, Alia-Klein NE, Goldstein RZ. Sex differences in cue-reactivity and reappraisal in heroin and cocaine use disorder — an fMRI study. Society for Neuroscience 2023; Washington, DC.

5. Huang Y, Butelman ER, **McClain N**, Hao Q, Dhamoon M, Robson P, Kaufman A, Goldstein RZ, Fayad Z, Alia-Klein N. Multi-modal imaging of carotid atherosclerotic risk in substance use disorders. Biomedical Engineering and Imaging Institute Symposium 2024; New York, NY.

6. **McClain, N***, Ceceli, AO*, Kronberg, G, Alia-Klein, N, Goldstein, RZ. (*equal contribution) Beyond self-report in characterizing drug addiction: Using behavior to inform treatment adherence. Friedman Brain Institute Retreat 2024; New York, NY.

7. Ceceli AO, King S, Drury K, **McClain N**, Gray J, Newcorn J, Schiller D, Alia-Klein N, & Goldstein RZ. Disrupting drug memories to diminish drug cue-reactivity using reconsolidation and methylphenidate in addiction: A double-blind randomized clinical trial in individuals with cocaine use disorder. American College of Neuropsychopharmacology 2024; Phoenix, AZ.

8. **McClain, N**, Drury, K, Kronberg, G, Alia-Klein, N, Goldstein, RZ. Reduced Synchronization of Neural Responses to a Drug-Related Movie in Opioid Use Disorder. Friedman Brain Institute Retreat 2025; New York, NY.

9. **McClain, N**, Kronberg, G, Drury, K, Alia-Klein, N, Goldstein, RZ. Menstrual Phase Effects on Brain State Dynamics during a Drug-Themed Movie in Cocaine Addiction. American College of Neuropsychopharmacology 2026; Nassau, Bahamas.

TECHNICAL SKILLS

- Computation: Proficient in Bash script, Python, E-Prime. Experience with MATLAB.
- Statistical analysis: Proficient in R, SPSS and GraphPad Prism.
- Neuroimaging data preprocessing and analysis: Proficient in FSL, Freesurfer, MRtrix. Experience with EEGLAB, Brainstorm, and FieldTrip.
- Study tool development: Proficient in REDCap and Qualtrics.

MENTORSHIP EXPERIENCE

Shraeya Santhapuram	MA student in Psychology at NYU
Christy Garlock	MA student in Psychology at NYU
Maggie Boros	Research Coordinator at NARC Lab
Rachel Drury	Research Coordinator at NARC Lab
Chloe Wong	MA student in Psychology at the New School

VOLUNTEERING

OnPoint NYC	July 2025-Present
Crown Heights Community Food Pantry	August 2025-Present